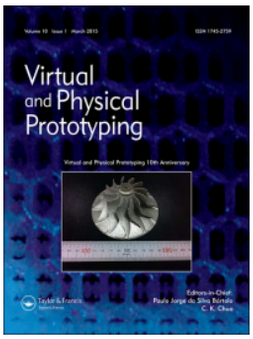

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3D Printing of chewable oral tablets using drug nanosuspension inks: an experimental and machine learning study

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ABSTRACT
3D printing enables precise control over tablet design and drug release, but challenges remain in optimising ink formulation, ensuring printability, and predicting final tablet properties. This study addresses the need for data-driven strategies in fabricating chewable tablets and tests the hypothesis that integrating rheology with machine learning (ML) enables predictive control over print quality and dosage form performance. We developed drug nanosuspension inks with varying water content (85–20 wt.%) and identified 40% as optimal, balancing shear-thinning behaviour, yield stress, and shear recovery for consistent extrusion. Analytical models predicted strut diameter (D) based on printing parameters—pressure (P), speed (v), and nozzle height (h)—but showed reduced accuracy under nonideal conditions. A Gradient Boosting Regressor (GBR) model improved predictions ($R^2 = 0.94$, RMSE = 52 µm) and enabled inverse prediction of printing settings for target dimensions. Tablets were characterised for drug stability, content uniformity, mechanical properties, and dissolution. XRD and DSC confirmed preserved GF crystallinity, and Raman and UV-VIS analysis demonstrated uniform drug distribution. Tablets had a stiffness of ~74 kPa, similar to commercial gummies, and dissolution rates were tunable via infill design. This work establishes a predictive, rheology-informed, and ML-enhanced framework for the scalable and customisable 3D printing of chewable oral tablets.

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Additive manufacturing; direct ink writing; pharmaceuticals; pediatric medicine; chewable tablets; machine learning, rheology

Introduction

Oral tablets remain the most widely used drug delivery method due to their cost-effectiveness, stability, and patient compliance [1,2]. However, conventional manufacturing methods offer limited flexibility in dosage customisation, often requiring tablet splitting, which introduces dosing errors and unpredictable drug release [3,4]. Additive manufacturing (AM) enables precision control over tablet geometry, porosity, and release kinetics, allowing for personalised dosing and multi-drug layering [5–7]. These advantages are particularly relevant for pediatric formulations, where off-label dosing, inter-individual variability, and patient compliance pose major challenges [8,9]. Pediatric adherence is influenced by taste, texture, tablet size, and swallowing difficulty, making chewable formulations more suitable for younger patients [10,11]. While gummy formulations appeal to children, they typically contain low API concentrations and suffer from stability issues during ink preparation and printing [12–14].

Several AM techniques have been explored for pharmaceutical manufacturing. Spritam® (levetiracetam), the first FDA-approved 3D-printed pharmaceutical, was developed using binder jetting, which produces a highly porous tablet that rapidly disperses in water [15,16]. While effective for rapid disintegration, binder jetting is not ideal for chewable formulations due to brittle tablet structures and reliance on chemical binders. Fused deposition modeling (FDM) is the most widely used due to its commercial availability, ease of processing, and filament-based drug loading [17–19]. However, FDM requires elevated temperatures (often exceeding 100°C), limiting its use for heat-sensitive APIs. Additionally, FDM-printed tablets tend to be dense and brittle, resulting in poor disintegration and mechanical properties unsuitable for chewable formulations. Similarly, light-based AM systems, such as stereolithography

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(SLA) and digital light processing (DLP), require resin baths containing photopolymerizable formulations, making them less suitable for tablet printing [5,20]. They often result in brittle tablet structures, require extensive post-processing to remove residual uncured resin, and pose risks of leaching unpolymerised material, raising safety concerns for pharmaceutical applications [21]. Moreover, both FDM and SLA/DLP generate substantial material waste during filament preparation or resin handling, further limiting their efficiency in pharmaceutical manufacturing.

Direct ink writing (DIW), an extrusion-based AM technique, offers a promising alternative by enabling room-temperature fabrication of viscoelastic drug inks, minimising API degradation risks while allowing precise control over tablet porosity, mechanical properties, and drug release [22–26]. DIW formulations can be optimised for chewability and flexibility while maintaining drug stability and controlled release. A variety of ink formulations have been explored for DIW of oral tablets, demonstrating the adaptability of extrusion-based AM techniques across different drug formulations. Hydroxypropyl methylcellulose (HPMC) is one of the most widely used binders and has been employed in lamotrigine [14], theophylline [27], carbamazepine [28], glipizide [29], spironolactone [9], and naftopidil formulations [30], often combined with polyvinylpyrrolidone (PVP) [28,29] or cyclodextrins [28] to enhance solubility and printability. Polyvinyl alcohol–polyethylene glycol (PVA-PEG) graft copolymers are another widely used class of binders, enabling room-temperature processing in aqueous media [25,31]. Sodium carboxymethylcellulose (Na-CMC) [32], microcrystalline cellulose (MCC) [9], dextrin [9], and croscarmellose sodium [33] have also been used as disintegrants to modify drug release. Gelatin is frequently utilised, leveraging gelatin's gelling properties for hydrogel-based soft formulations [13,14,34]. The most commonly used solvents include water [14,24,25,27,31] and ethanol solution (in water) [26,28,29,32], in some cases with addition of organic solvents (e.g. dimethyl sulfoxide, acetone, methanol) [23,35], ensuring compatibility with temperature-sensitive APIs, while soybean oil-based formulations (e.g. amlodipine besylate, fenofibrate) provide additional control over drug solubility and release [33,36].

Despite its promise, most direct ink writing (DIW) approaches for pharmaceutical applications face critical limitations, including reliance on elevated printing temperatures and organic solvents (e.g. ethanol), which restrict compatibility with thermolabile drugs and patient-friendly dosage forms. Additionally, many studies focus on Biopharmaceutics Classification System (BCS) Class I drugs with high solubility and permeability, overlooking the more pressing formulation challenges associated with BCS Class II drugs. Even when low-solubility drugs are studied, dissolution testing is often performed under pH-adjusted or surfactant-rich conditions that artificially enhance solubility, reducing the physiological relevance of the results.

To address these gaps, this study develops temperature-independent, solvent-free nanosuspension inks for DIW of soft tablets containing BCS Class II drugs, using griseofulvin (GF)—a poorly water-soluble antifungal drug with low oral bioavailability and a history of complex formulation strategies—as a model system. We utilise wet-milled GF nanosuspensions and perform all dissolution studies in 3 g/L SDS medium, which mimics intestinal fluid conditions without artificially boosting solubility beyond biorelevant levels. Through detailed rheological characterisation, printability assessments, and machine learning (ML)-driven prediction of printing outcomes, we establish a framework that links ink rheology to print quality and drug release behaviour. This work also explores the potential of DIW for customised oral dosage forms, including chewable tablets and pediatric-friendly designs. By integrating experimental and computational tools, our approach advances DIW as a scalable and clinically relevant platform for the formulation of poorly soluble drugs.

Results and discussion

Formulation of drug nanosuspension inks

Drug nanosuspensions were prepared by wet stirred media milling (WSMM) of griseofulvin (GF) and sodium starch glycolate (SSG) particles (Figure 1A) in deionized (DI) water, with sodium dodecyl sulfate (SDS) and hydroxypropyl cellulose (HPC) as stabilisers (Table 1). WSMM was choosen for this study as it is the most widely used and industry-relavant milling process for manufacturing drug nanoparticles [37–39]. GF, an antifungal drug, served as the model Biopharmaceutics Classification System (BCS) Class II active pharmaceutical ingredient (API) [40]. SSG particles were used as an insoluble but swellable disintegrant in DI water. The size reduction in WSMM occurs through mechanically driven particle breakage, which is often hindered by strong interparticle attractions (e.g. van der Waals or hydrophobic forces) that lead to aggregation [41,42]. To prevent paticle aggregation, an electrostreic stabilisation approach was employed, where steric stabilisation was provided by nonionic polymer HPC, and electrostatic stabilisation was imparted by SDS, an anionic surfactant [39,43,44]. In this method, HPC and SDS adsorped onto the particle surfaces in solution, working together to stabilise the drug nanosuspension ink formulation (Figure 1B) [37]. The formulation presented in Table 1 was designed based on prior work from the Bilgili lab [45–49], which

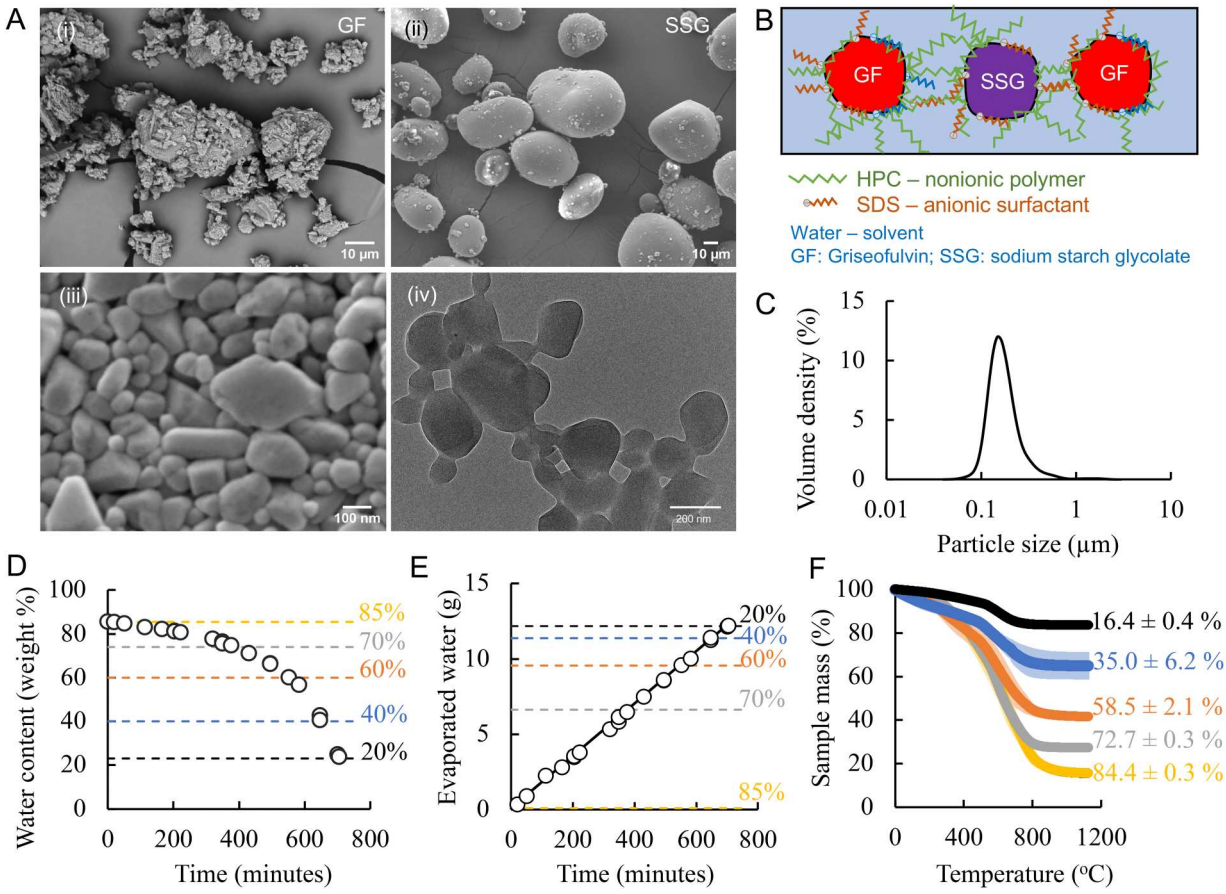

Figure 1. Formulation and characterisation of nanosuspension inks: (A) SEM images of (i) as recieved griseofulvin (GF), (ii) sodium starch glycolate (SSG), and (iii) milled nanosuspension, and (iv) TEM image of the nanosuspension. Scale bars: 10 micron (i-ii), 100 nm (iii) and 200 nm (iv). (B) Schematic of drug nanosuspension ink showing GF and SSG nanoparticles surrounded by nonionic polymer (hydroxypropyl cellulose, HPC) and anionic surfactant (sodium dodecyl sulfate, SDS). (C) Particle size distribution (vol.%) of the milled nanosuspension. (D) Water content (wt.%) as a function of incubation time at 35°C during preparation of various ink formulations including 85% (as-milled), 70%, 60%, 40%, and 20% water. (E) Mass of water evaporated as a function of time corresponding to (D). Solid line shows the linear fit ($y = 0.0173x$; $R^2 = 1$. F) TGA plots showing % mass loss with temperature. Legends represent the calculated % water (mean ± std) in each formulation.

has extensively investigated nanosuspension-based drug delivery systems and established effective protocols for stabilising poorly water-soluble drugs.

The milling parameters—operation at 3200 rpm for 90 minutes with 50 mL of yttrium-stabilized zirconia beads in an 80 mL chamber—were selected based on prior studies that established effective protocols for particle size reduction and stabilisation in similar systems [45,46,48]. These studies demonstrated that the selected milling conditions provide consistent nanoparticle generation and reproducibility. Although the specific formulation in this study is novel, it builds upon previous work in the Bilgili lab. For example, Bilgili et al. optimised HPC concentrations for 10 wt% drug with 0.5 wt% SDS [46], while Azad et al. investigated the role of 2 wt% SSG in nanosuspensions [47]. The inclusion of SSG in our formulation was similarly motivated by its demonstrated benefits in enhancing drug delivery performance, including improved disintegration and bioavailability in in vivo models [50]. Additionally, XRD and DSC analyses were conducted to confirm that the milling process did not alter the crystalline structure of GF, thereby preserving its physical stability. The results of these analyses are discussed in a later section of the manuscript.

Table 1. Composition of the drug nanosuspension ink fomulations.

API GF (g)	Disintegrant SSG (g)	Ionic surfactant SDS (g)	Nonionic surfactant HPC (g)	Solvent DI water (mL)	Water (wt.%)
10	1.5	0.5	5	100	85
10	1.5	0.5	5	40	70
10	1.5	0.5	5	26	60
10	1.5	0.5	5	11	40
10	1.5	0.5	5	4	20

API: Active pharmaceutical ingriedient; GF: Griseofulvin; SSG: Sodium starch glycolate; HPC: Hydroxypropyl cellulose

The average particle size of the nanosuspension was measured as 185 ± 2 nm (d10 = 108 ± 10 nm, d50 = 146 ± 1 nm, d90 = 199 ± 2 nm), indicating a relatively uniform size distribution (Figure 1C). SEM and TEM images revealed that the particle had a smooth morphology (Figure 1A). Image analysis of TEM micrographs provided a mean particle size of 139 ± 10 nm and a mean aspect ratio of 1.41 ± 0.34. In addition to as-milled formulation (containing 85% DI water), four additional drug nanosuspension inks were prepared by adjusting the water content to 70%, 60%, 40% and 20% via controlled evaporation at 35°C (Figure 1D). The water mass rate was calculated as 0.0173 g·min^{-1} (Figure 1E). The final water concentration of each ink formulation was confirmed using TGA (Figure 1F).

Rheological properties of drug nanosuspension inks

The changes in ink viscosity (η) as a function of shear rate ($\dot{\gamma}$) for formulations with varying water content (85-20 wt.%) are shown in Figure 2A. All ink formulations exhibited shear-thinning behaviour, as indicated by the significant decrease in η with increasing $\dot{\gamma}$. The Hershel-Bulkley (HB) model [51] was used to describe the viscoelastic behaviour of the inks:

$$\tau = \tau_0 + K\dot{\gamma}^n \quad (1)$$

where τ is the applied shear stress, τ_0 is the dynamic yield stress, K is the consistency index, and n is the flow index [51]. In this model, τ_0 represents the critical stress required to initiate ink flow, with $n < 1$ indicating shear thinning behaviour. If $n = 1$ and $\tau_0 = 0$, the ink behaves as a Newtonian fluid. To analyse the data, τ was plotted as a function of $\dot{\gamma}$ in Figure 2B. The HB model was primarily fitted to the low shear rate region to accurately capture τ_0, which is critical for assessing printability. While the model parameters were extracted from the low shear regime, the overall flow behaviour remained consistent across the shear rate range relevant to extrusion-based printing. However, for ink formulations with lower water content, we observed experimental limitations at higher shear rates—likely due to increased viscosity and edge effects. These limitations affected data quality at the highest shear rates, but the inks still exhibited a continuous shear-thinning trend, validating the applicability of the model for these formulations

As summarised in Table 2, all ink formulations showed shear thinning behaviour ($n < 1$), with the degree of shear thinning increasing as water content decreases. This trend is consistent with the changes in zero-shear viscosity (η_0) and average shear viscosity (η_{avg}) as a function of water content (Table 3). Specifically, η_0 was obtained from low shear rate region ($\dot{\gamma} < 0.01$ s^{-1}) and η_{avg} represents the average viscosity within the range $100 < \dot{\gamma} < 300$ s^{-1}, corresponding to the shear rates experienced during printing (see Experimental Section).

The increase in η_0 and reduction in flow index (n) with decreasing water content reflects enhanced particle–particle interactions and the formation of a denser microstructure, which is consistent with prior observations in concentrated suspensions [52]. In our nanosuspension inks, these interactions are influenced not only by the total solid content but also by the properties of the polymeric stabiliser. Hydroxypropyl cellulose (HPC), a nonionic, water-soluble polymer used in the formulation, adsorbs onto the surfaces of griseofulvin and sodium starch glycolate particles during wet stirred media milling. This adsorption provides steric stabilisation that suppresses aggregation and promotes dispersion. At higher solid contents (i.e. lower water content), the reduced interparticle spacing increases the frequency and strength of interactions between polymer-coated particles, contributing to increased viscoelasticity and higher yield stress. While a single polymer formulation was employed across all conditions, these concentration-dependent effects of HPC on interparticle dynamics play a critical role in shaping the rheological behaviour of the inks.

Oscillatory tests revealed that the shear modulus increased with frequency, with minimal change observed at 1 Hz (Figure 2C). To investigate the transition between elastic and viscous behaviour, shear strain sweeps (at 1 Hz) were conducted, measuring the change in elastic (G') and viscous (G'') moduli with applied shear strain (Figure 2D). At low strain, all formulations except for 85% ink behaved as elastic solids ($G' > G''$). At a critical strain ($G' = G''$, Figure 2D), the formulations transitioned to viscous liquid-like behaviour, defining the yield stress (σ_y) for each ink formulation (Table 3).

The σ_y represents the minimum shear stress required to initiate ink flow during printing. Compared to τ_0, which is derived from the HB model, σ_y is directly measured from oscillatory tests and is frequency dependent [53]. Various methods have been reported in the literature for estimating σ_y, including: (i) crossover point ($G' = G''$) [54,55] used in this study, (ii) G' from the linear viscoelastic region (G'_0) [56–58], or (iii) the onset of G' deflection [59,60]. The measured σ_y values followed the same trend as τ_0, increasing with decreasing water content, with $\tau_0 > \sigma_y$, in agreement with literature [52,53].

The ratio of G''_0/G'_0 (or tan δ) is a critical parameter indicating the ink's state at rest. A G''_0/G'_0 value grater

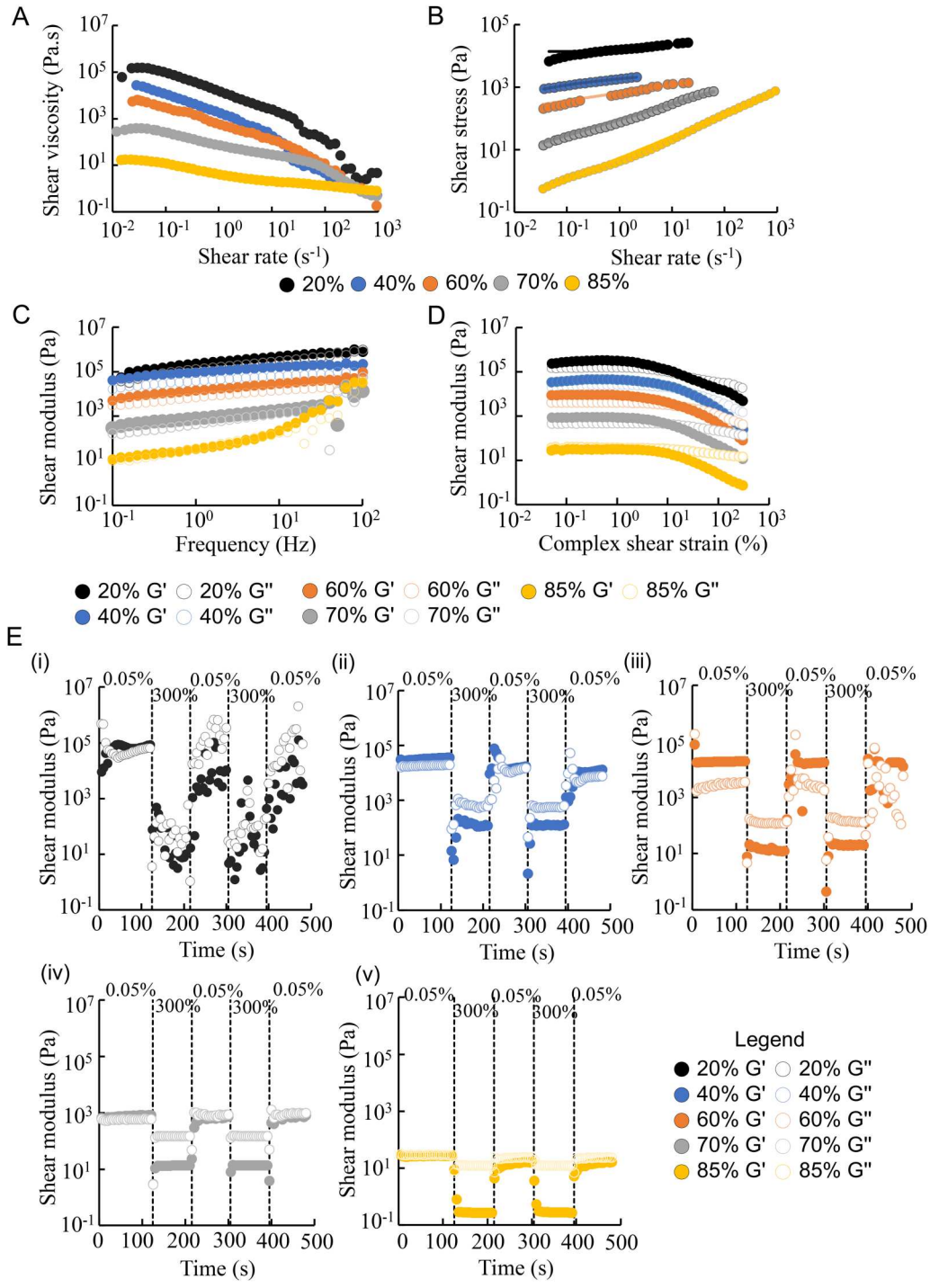

Figure 2. Rheology of ink formulations (20%, 40%, 60%, 70%, and 85%). (A) Shear viscosity with shear rate. (B) Shear stress with shear rate (solid lines represent Hershel Bulkley model fits, Table 2). (C) Shear modulus (elastic (G') and viscous (G'') modulus) with frequency. (D) Shear modulus with shear strain for each formulation. (E) Shear modulus with time at repetative low (0.05%) and high (300%) strain.

than 1 signifies a fluid-like behaviour, meaning the ink dissipates a significant amount of energy when sheared, while a value less than 1 indicates a solid-like behaviour, where elastic properties dominate. As shown in Table 3, for 20–70% ink formulations, G''_0/G'_0 ranged from 0.5 to 0.6, indicating solid-like behaviour at rest, whereas for 85% ink, G''_0/G'_0 was ~1.3, demonstrating a liquid-like behaviour. Maintaining a solid-like behaviour at rest is crucial for extrusion-based 3D printing as it prevents ink from dripping out of the nozzle before extrusion, ensuring printing precision and reducing material waste. It also enhances shape retention

Table 2. Summary of the HB model fitting parameters including yield stress (τ_0), consistency index (K) and flow index (n). S denotes the standard error of regression, and R^2 is the coefficient of determination.

Ink	τ_0 (Pa)	K (Pa s^n)	n	S (Pa)	R^2
20%	9668 ± 849	5706 ± 191	0.389 ± 0.015	757	0.98
40%	240 ± 37	1614 ± 9	0.244 ± 0.005	28.4	0.99
60%	160 ± 10	471 ± 19	0.354 ± 0.019	64.3	0.97
70%	9.3 ± 0.9	77 ± 3	0.566 ± 0.012	17.6	0.99
85%	0.6 ± 0.1	3.8 ± 0.1	0.774 ± 0.002	2.18	1.00

and structural integrity post-deposition, preventing collapse and enabling accurate layer-by-layer fabrication.

Shear-thinning behaviour facilitates the extrusion of highly concentrated nanosuspension inks, but shear recovery is essential for maintaining the shape accuracy of printed structures [61,62]. Shear recovery is typically assessed by step shear strain tests [63,64], where the ink is subjected to a low shear strain (0.05%) that simulates steady-state conditions before or after printing, followed by exposure to a high shear strain (300%) that significantly exceeds the yield strain (Figure 2E). For effective extrusion-based printing, a yielding fluid should exhibit rapid shear recovery to ensure self-supporting layers with minimal sagging [65]. However, to prevent nozzle clogging, σ_y should not exceed 4,000 Pa to prevent clogging [27].

The results in Figure 2E show that all formulations except 85% ink exhibited solid-like behaviour ($G' > G''$) at low shear strain, transitioning to liquid-like behaviour ($G'' > G'$) at high shear strain (300%). Shear recovery times were relatively similar across formulations, with values of approximately 40 s for 40% ink, 35 s for 60% ink, and 40 s for 70% ink. The 85% ink exhibited liquid-like behaviour in both strain regimes, while the 20% ink showed strong fluctuations, indicating film fracture or slip conditions [66]. These findings provide insights into the rheological behaviour of the inks, which will be further correlated with shape retention and printability in the following section.

Printibility of inks

Assessing the formation and stacking of printed struts (lines) is essential for evaluating and predicting printability and shape accuracy in extrusion-based 3D printing [67–69]. In this study, semi-quantitative assessment methods were used based on the shape fidelity of square pores and the self-supporting ability of disc-shaped solid tablets [68,69]. The printing pressure and speed were optimised (~400 µm offset) by printing continuous (~550 µm) struts (Figure 3A).

Initially, it was found that 20% ink was not extrudable due to its high viscosity. The 3D-printed grid patterns from the 40% ink maintained their pore shape and size, whereas pores became circular and smaller for 60% and 70% inks and completely disappeared for 85% ink, indicating coalescence of neighbouring struts (Figure 3B and 3C). The ideal grid design should exhibit a square pore profile (in the x-y plane), and a printability index (P_r), defined as the ratio of the printed pore area to the design pore area, was used to assess printability. A P_r value of 1 represents a perfectly retained square pore, while $P_r < 1$ indicates pore shrinkage and rounding, and $P_r > 1$ signifies irregular pore deformation (Figure 3B). Similarly, 3D-printed tablets using 40% ink formulation retained their shape, as indicated by 90° contact angle (Figure 3D and 3E). Contact angles decreased with increasing water content, indicating sagging (spreading) due to surface tension and gravity [53]. Tablets printed using 85% ink could not support their weight and completely wetted the printed surface.

The observed differences in printability can be directly correlated with rheological properties, particularly shear recovery and yield stress (σ_y). While shear recovery times were similar (~35–40s) for the 40%, 60%, and 70% inks, the elastic modulus (G'_0) and yield stress (σ_y) progressively decreased with increasing water content. A sufficiently high σ_y is crucial to maintain structural integrity and prevent sagging, but it must also remain below 4,000 Pa to avoid clogging. The 40% ink formulation achieved the optimal balance, exhibiting a strong enough structure to retain pore geometry and tablet shape, whereas 60% and 70% inks suffered from excessive spreading due to lower yield stress. The 85% ink behaved like a liquid in all strain regimes, preventing any meaningful structure formation, while 20% ink was too viscous to be extrudable.

Based on our results, 40% ink formulation was selected as the optimum formulation and used in the

Table 3. Summary of rheological parameters including zero shear viscosity (η_0), average shear viscosity during printing (η_{avg}), viscous modulus (G''_0) and elastic moduls (G'_0) from linear viscoelastic region, and static yield stress (σ_y) obtained at ($G' = G''$).

Ink	η_0 (Pa·s)	η_{avg} (Pa·s)	G''_0 (Pa)	G'_0 (Pa)	G''_0/G'_0	σ_y (Pa)
20%	154,625 ± 3,200	56.7 ± 45.6	148,736 ± 3,288	300,407 ± 29,292	0.50	18,650
40%	102,890 ± 7,153	13.3 ± 6.8	23,122 ± 1,915	42,216 ± 4,069	0.55	3,975
60%	2,976 ± 566	2.9 ± 1.7	4,141 ± 14	9,081 ± 159	0.46	724
70%	341 ± 32	3.9 ± 2.0	469 ± 6	877 ± 9	0.53	62
85%	60 ± 3	1.4 ± 0.1	38 ± 2	30 ± 1	1.26	0

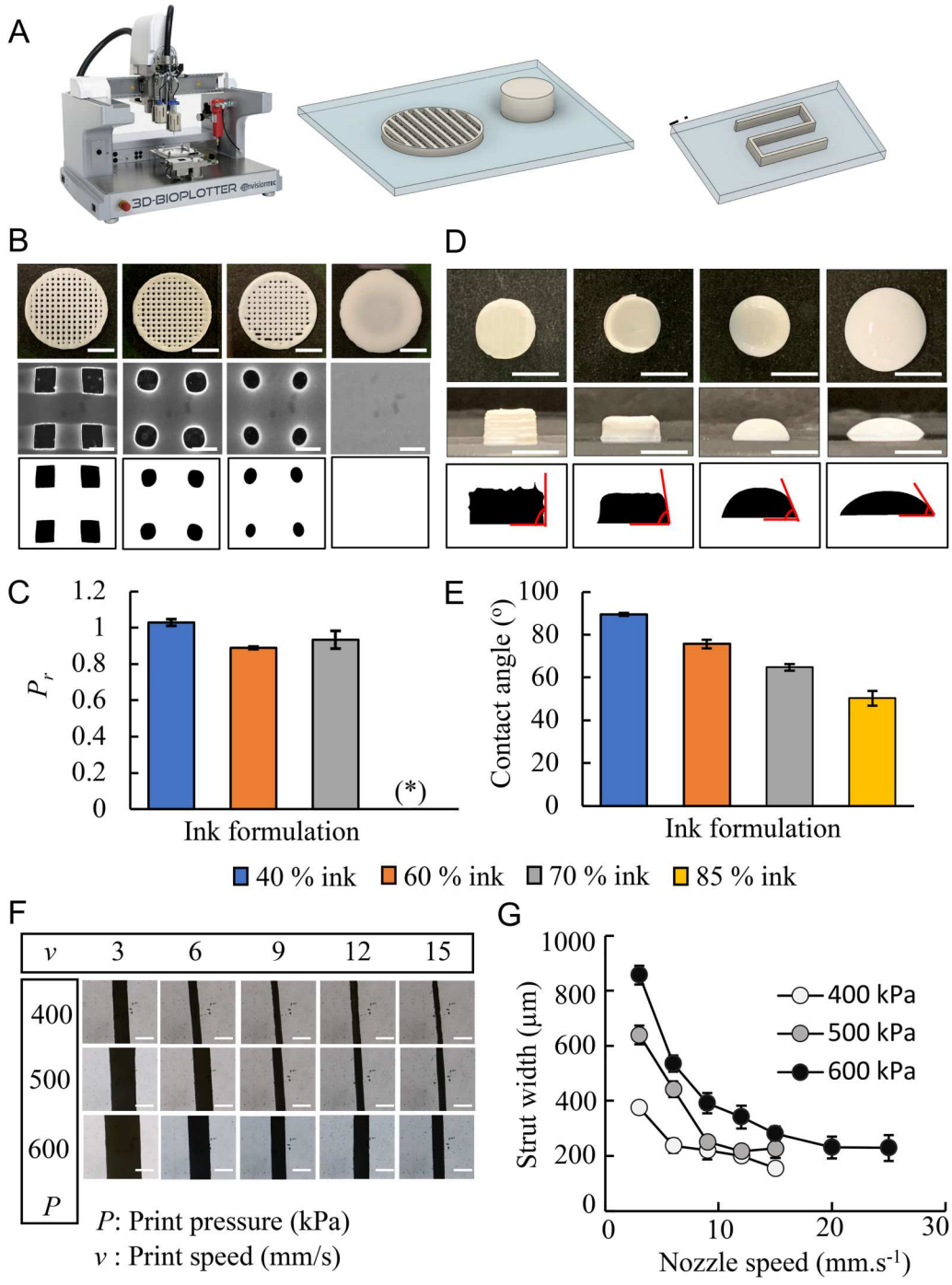

Figure 3. Characterisation of the printability of ink formulations (40%, 60%, 70%, and 85%). (A) Image of the 3D printer and digital designs for the grid, tablet, and continuous line structures. (B) Printed grid designs, including photographs of the scaffolds (top) (scale bars = 5 mm), optical images of the pores (middle) (scale bars = 500 µm), and thresholded images of pores (bottom). (C) Printability index (Pr) plotted for each ink formulation. (*) Indicates that the 85% ink fully collapsed. (D) Printed solid discs, including top-view images (top) (scale bars = 5 mm), cross-sectional images (middle) (scale bars = 5 mm), and thresholded images used for contact angle calculation (bottom). (E) Plot of the corresponding contact angle values for each ink formulation. (F) Optical images of printed struts for the 40% ink under varying print pressures (400-600 kPa) and print speeds (3-15 mm/s) (scale bars = 500 µm). (G) Measured strut width corresponding to the conditions in (F). Data are presented mean ± std. ($n \geq 3$).

remainder of this study. To assess the effect of print pressure (*P*) and speed (*v*) on strut width or diameter (*D*), a continuous line was printed at ~400 µm nozzle offset (for 400 µm nozzle-diameter), and *D* was measured (Figure 3F and 3G). Under optimal printing conditions, *D* describes the diameter of a cylindrical

strut, which may deviate from cylindrical geometry due to lateral compression by the nozzle tip caused by excessive ink flow (at high print pressures or low speeds) or by insufficient flow (at low print pressures or high speeds). Notably, the use of the 40% ink formulation eliminated sagging, which otherwise lead to an increased strut width. Overall, increasing P led to larger struts, with widths increasing as v decreased (~850 µm at 600 kPa and 3 mm/s). Fine struts (180–200 µm) were achieved at 400 kPa for speeds ranging from 6 to 15 mm/s.

Connecting rheology with printability via analytical modelling and machine learning

Analytical models can be applied to predict optimal printing parameters, such as P, v, and nozzle diameter, using rheology data, including viscosity and shear thinning behaviour. D is correlated with volumetric flow rate of the ink (Q) and v, as shown below [70]:

$$D = \sqrt{\frac{4Q}{\pi v}} \tag{2}$$

where Q can be either directly measured Q_E or calculated Q_C using rheology data. Q_C accounts for both shear and slip components ($Q_C = Q_{shear} + Q_{slip}$) [70], which can be described for a Herschel–Bulkley fluid extruded through a tapered needle using the following equations [70–72]:

$$Q_{shear} = \frac{\pi r_i^3 r_0^3}{4}\left[\frac{3n\tan\theta_c\left(\Delta P + \frac{2\tau_0}{\tan\theta_c} Ln\left(\frac{r_i}{r_0}\right)\right.}{2K(r_i^{3n} - r_0^{3n})}\right]^{\frac{1}{n}} \tag{3}$$

$$Q_{slip} = \frac{\pi\delta_t r_o^2}{K^{1/n}}\left[\tau_o + \frac{3n\ \Delta P\ \tan\theta_C}{2\left(1 - \left(\frac{r_o}{r_i}\right)^{3n}\right)}\right]^{1/n} \tag{4}$$

Here, θ_c is the half cone angle ($\tan\theta_c = (r_i - r_0)/L$, where L is the tapered nozzle length), r_i and r_0 are the inlet and outlet radii of the tapered needle, and δ_t is the thickness of the slip layer (~15% of the mean particle size of the suspended particles) [73]. The pressure drop through the print needle is denoted as ΔP, which account for the applied print pressure (P), atmospheric pressure (P_a), and additional pressure drop (P_c) due to frictional losses for non-Newtonian fluids and flow through fittings [74]. Both equations use τ_0, n, and K values obtained from the HB model fit (Figure 2B).

The volumetric flow rate, Q, increased with P, and Q_C values showed good agreement with Q_E (Figure 4A, Table 4). The experimental strut diameter (D_E) also closely matched the predicted values (D_M), confirming the accuracy of the model, except at low print speeds for $P = 500$ ($v < 4$ mm/s) and 600 kPa ($v < 7$ mm/s) (Figure 4B). These deviations are attributed to print nozzle compressing the extruded ink when D exceeds nozzle offset (h). Increasing h to 500 µm eliminated this issue in most cases, except at $v = 3$ mm/s for 600 kPa. Further increasing h to 700 µm resulted in significantly lower D_M values, as the gap became too large to be fully filled with the extruded ink. A similar behaviour was observed for 400 kPa at a 500 µm nozzle offset. Overall, the model accurately predicted D as long as h was appropriately adjusted based on P and v. However, under nonideal conditions–specifically during excessive flow (high P and slow v) or insufficient flow (low P and faster v)–the model failed to predict D accurately (Figure 4C). These deviations are inherently linked to the model's assumption of a perfectly cylindrical strut with a circular cross-section. In practice, deviations from this ideal geometry arise due to the nozzle-to-substrate offset: if h is too small, struts may be flattened or squeezed [75–77]; if h is too large, the struts may not fully bridge the gap, leading to breakup or thinning. While these effects could theoretically be incorporated into the model by accounting for h-dependent deformation, doing so would significantly increase model complexity and likely require empirical calibration. As such, the current model is best suited for predicting strut geometry under near-ideal extrusion conditions, with the noted limitations under extreme parameter combinations.

To address these challenges, we hypothized that machine learning (ML) could predict D not only under ideal conditions but also for non-ideal conditions. Additionally, ML could enable inverse predictions to determine optimal combinations of P, v, and h for a target D.

In the ML approach, principal component analysis (PCA), K-fold cross-validation, and data scaling were integrated to improve model efficiency, accuracy, and reliability. PCA reduced data dimensionality by capturing key features and eliminating noise and irrelevant variables, thereby enhancing prediction accuracy [78]. K-fold cross-validation ensured robust model evaluation by preventing overfitting and improving generalizability, as it trained and tested on different data subsets [79]. Data scaling was applied to standardise features, ensuring equal weighting in the model and improving convergence, particularly for models sensitive to feature magnitudes, such as Gradient Boosting and K-Neighbors Regressor models [80]. In the ML analysis, D was set as the response variable, while P, v, and h were assigned as predictors. Among the regression models tested–including Linear Regressor (LR), Random Forest

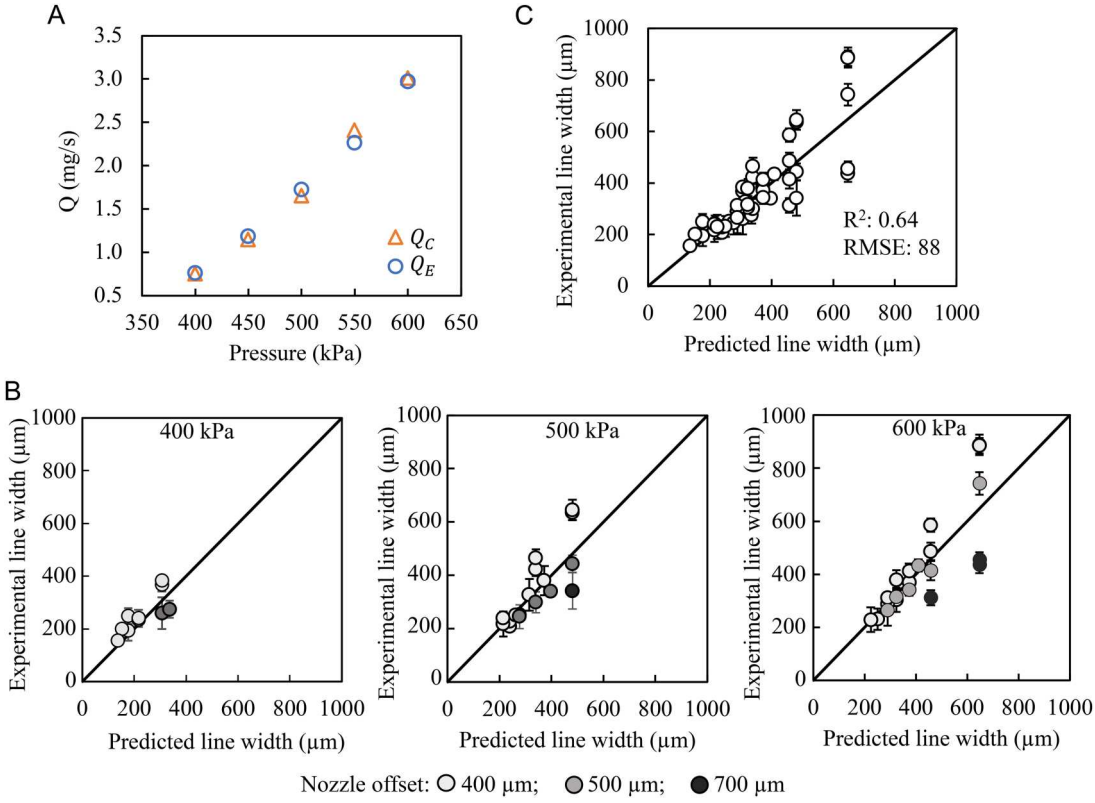

Figure 4. (A) Change in ink flow rate (calculated (Q_C) and experimentally measured (Q_E)) with print pressure (P). (B) Experimentally measured line width plotted against predicted line width for $P = 400$, 500, and 600 kPa and nozzle offset = 400, 500, and 700 μm. (C) Experimentally measured line width plotted against predicted values for all conditions.

Regressor (RFR), Gradient Boosting Regressor (GBR), and K-Neighbors Regressor (KNR)– GBR demonstrated the strongest predictive accuracy, as indicated by a low RMSE (52) and high R^2 (0.94) (Figure 5A-5D). The superior performance of GBR is attributed to its ensemble learning approach, which effectively reduces bias and variance in the predictions [81]. Although RFR is also an ensemble method, it was less effective, likely due to overfitting or insufficient depth in capturing predictor interactions [82]. In contrast, LR failed to capture the relationships between predictors due to their clearly nonlinear nature. Overall, GBR was determined to be the most reliable model for predicting D and was selected for the inverse prediction of printing parameters (P, v, and h) to achieve target D values.

Table 4. Summary of the analytical modelling data, including print pressure (P), pressure drop (ΔP), calculated flow rate (Q_C), and experimentally measured flow rate (Q_E). Q_C consists of shear (Q_{shear}) and slip (Q_{slip}) components.

P (kPa)	ΔP (kPa)	Q_{shear} (mg/s)	Q_{slip} (mg/s)	Q_C (mg/s)	Q_E (mg/s)	*PE (%)
400	218	0.752	0.236 x10^{-3}	0.752	0.763	1.3485
450	251	1.148	0.373 x10^{-3}	1.148	1.183	2.9871
500	283	1.651	0.550 x10^{-3}	1.652	1.725	4.2376
560	320	2.408	0.822 x10^{-3}	2.409	2.263	6.4568
600	344	3.008	1.039 x10^{-3}	3.009	2.970	1.3039

*Percentage error $[((Q_E - Q_C)/Q_E) \times 100]$ is classified as 0–5% (high accuracy), 5–10% (acceptable accuracy), 10–20% (moderate error), and >20% (high error).

The accuracy of the inverse predictions was verified by comparing the target D_T with the actual D_A obtained experimentally using the predicted printing parameters (P, v, and h) (Table 5). As shown in Figure 5E, the predictions were highly accurate (RMSE = 26 and $R^2 = 0.95$). Additionally, the model provided multiple parameter combinations to achieve the target D_T, offering flexibility in selecting optimal printing conditions. Table 6 summarises the combinations of printing parameters and corresponding actual D_A for target $D_T = 300$, 350, and 400 μm. The agreement between the target and actual values in Figure 5F further confirms the model's accuracy. Overall, ML proved to be an effective and practical tool for predicting D for given printing parameters and for determining optimal printing parameters for a desired D.

In this study, our ML model focused on printing parameters and assumed that the material (or ink) properties remained consistent across different batches, which was confirmed by measuring the ink viscosity for each batch. In the future work, we plan to incorporate variations in ink viscosity or composition into the

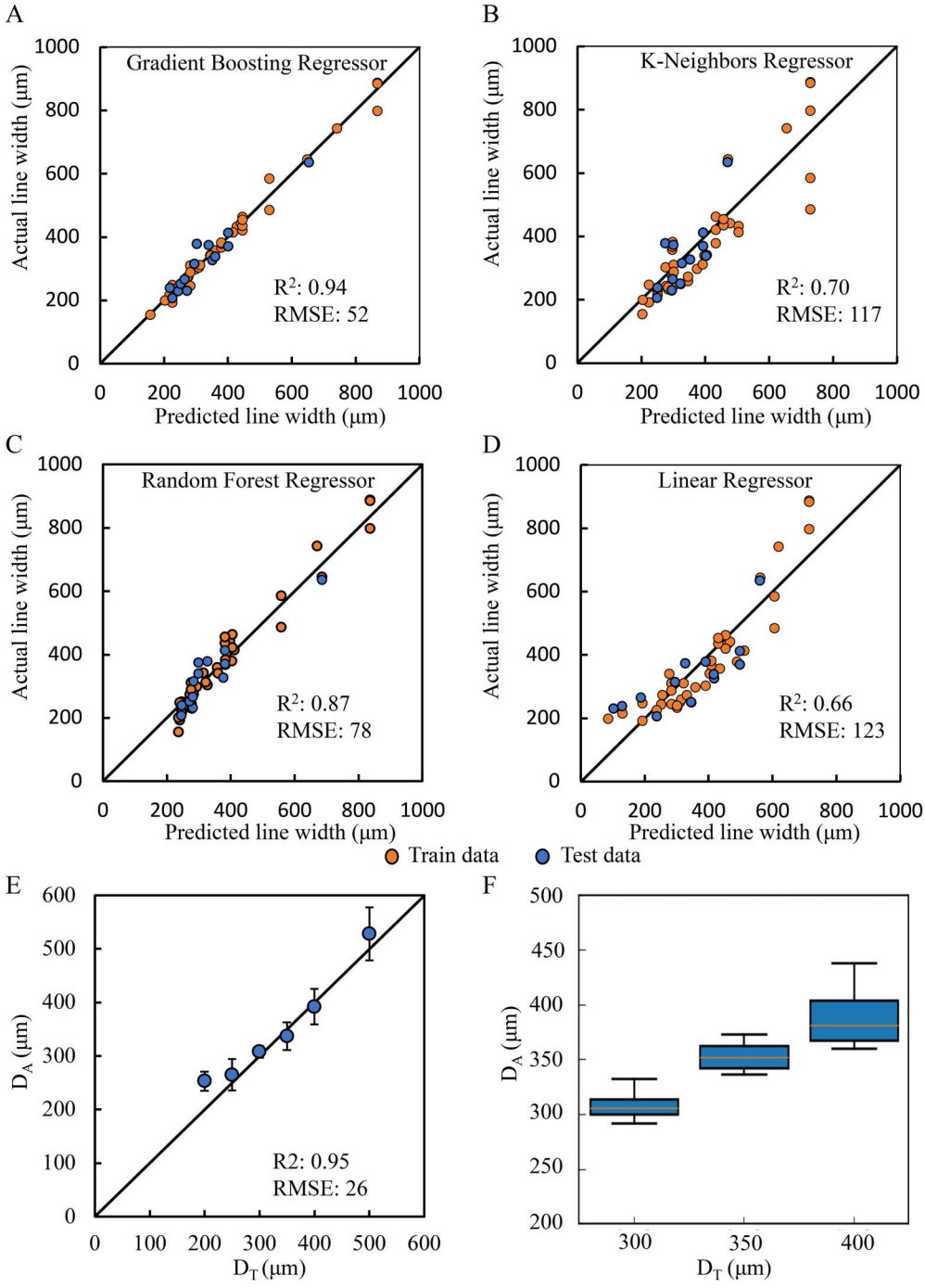

Figure 5. Summary of machine learning outcomes: (A-D) Model efficiency and validation for Gradient Boosting, K-Neigbors, Random Forest, and Linear Regressor models. (E-F) Inverse prediction of printing parameters from the target line width using the General Bossting Regressor model. (E) Actual line width measured for struts printed using a single set of predicted print parameters. (F) Actual line width measured for struts printed using multiple sets of predicted print parameters.

model to further enhance its predictive capabilities and extend its applications to ink formulation design.

3D printing and characterisation of tablets

Drug nanosuspension inks containing 40% water were used to 3D print oral tablets in the form of discs with 6 mm diameter (100% and 50% infill) and 9 mm diameter (50% infill) (Figure 6A). Micro-CT imaging confirmed the uniformity and structural integrity of the printed tablets (Figure 6B). A comprehensive characterisation was conducted to evaluate drug stability, content uniformity, mechanical behaviour, porosity, and drug release profiles. Tablets produced via direct compression of

Table 5. Prediction of printing parameters nozzle offset (h), print pressure (P) and print speed (v) for target D (D_T).

D_T (μm)	h (μm)	P (kPa)	v (mm/s)	D_A
500	433	547	4.5	528 ± 49
400	407	467	7.9	392 ± 33
350	433	433	4.8	337 ± 26
300	430	426	4.7	308 ± 11
250	407	377	5.1	265 ± 29
200	417	407	6.8	253 ± 18

Table 6. Prediction of multiple sets of printing parameters nozzle offset (h), print pressure (P) and print speed (v) for target D (D_T).

D_T (μm)	h (μm)	P (kPa)	v (mm/s)	D_A
300	380	525	6.2	332 ± 26
	430	426	4.7	308 ± 11
	421	467	8.5	302 ± 18
	410	435	5.8	292 ± 35
350	433	433	4.8	337 ± 26
	397	448	5.8	373 ± 34
	410	454	5.3	344 ± 26
	380	525	6.2	359 ± 6
400	407	467	4.5	392 ± 33
	386	478	5.8	370 ± 9
	381	455	3.7	360 ± 6
	403	513	7.1	438 ± 26

the powder mixture (PM) and pure GF powder served as control groups.

Crystallinity

The crystallinity of GF was monitored to ensure that the printing process did not induce a phase change. Phase transformations in APIs during manufacturing can significantly impact drug efficacy, shelf-life, and bioavailability [83,84]. For instance, 3D printing at elevated temperatures can induce crystallinity loss, sometimes leading to an amorphous state [14,85,86]. However, XRD and DSC profiles of 3D-printed tablets were similar to those of PM and GF powder, confirming that GF retained its crystalline form post-printing (Figure 6C, D). These analyses were conducted on the 40% ink formulation after complete drying, as it represented the most extensively processed and printed sample. Notably, amorphous GF exhibits a glass transition at 88.9°C and melts at 217°C [87]. Additionally, it undergoes a bimodal crystallization event at ~127°C [88], which was absent in the DSC profile, further confirming the preservation of crystallinity (Figure D).

Processing and operator effects on tablet uniformity

Regulation and quality control are critical for public trust in AM technologies [89]. A well-documented concern in the literature is the lack of standardised quality control metrics in AM, which extends beyond traditional mass-based assessments of compressed tablets [13,90,91]. While AM offers significant advantages–such as customised dosages, multi-API combinations, and patient-specific formulations– it also represents challenges regarding regulatory oversight and operator-dependent variability in production.

To assess whether the ink formulation or printing process compromised intra-tablet homogeneity, Raman mapping and UV-VIS detection were performed. Raman mapping of 3D-printed tablets indicated relatively uniform distributions of GF, HPC, SDS, and SSG across the bottom, centre, and top print layers (Figure 7A). Additionally, UV-VIS dissolution tests of the top, centre, and bottom print layers showed no significant variation in GF content (~4.2 mg per layer, corresponding to ~50 wt.%) among layers (Figure 7A). These findings confirm that the ink formulation remained stable, and the 3D printing process successfully produced tablets with high content uniformity.

We further investigated operator-dependent variations in tablet quality and content uniformity (Figure 7B). Three operators with varying levels of expertise were selected: Opr. 1 – An inexperienced undergraduate student; Opr. 2 – A graduate student with basic training; Opr. 3 – A senior graduate student with extensive 3D printing expertise. Each operator was provided with a 40% ink formulation and followed the same procedure: loading the syringe, calibrating the printer, printing the tablets, and collecting them for analysis. Results showed no statistically significant differences in tablet weight, dried tablet weight, or total GF content (32.8 ± 3.4 mg, n = 12) across tablets printed by different operators.

These findings highlight the robustness and reproducibility of the 3D printing process, demonstrating that even with varying levels of operator expertise, tablet quality and content uniformity remain consistent. When minor deviations in viscosity were observed, targeted adjustments through controlled drying or dilution allowed us to restore optimal flow behaviour, reinforcing the process's adaptability. Together, these practices support the establishment of additive manufacturing as a dependable and scalable platform for pharmaceutical production, enabling consistent drug dosing and compliance with regulatory standards.

Mechanical behaviour

The mechanical behaviour of an oral tablet is critical for safety, performance, and patient acceptance [92]. For instance, chewable tablets are significantly softer than

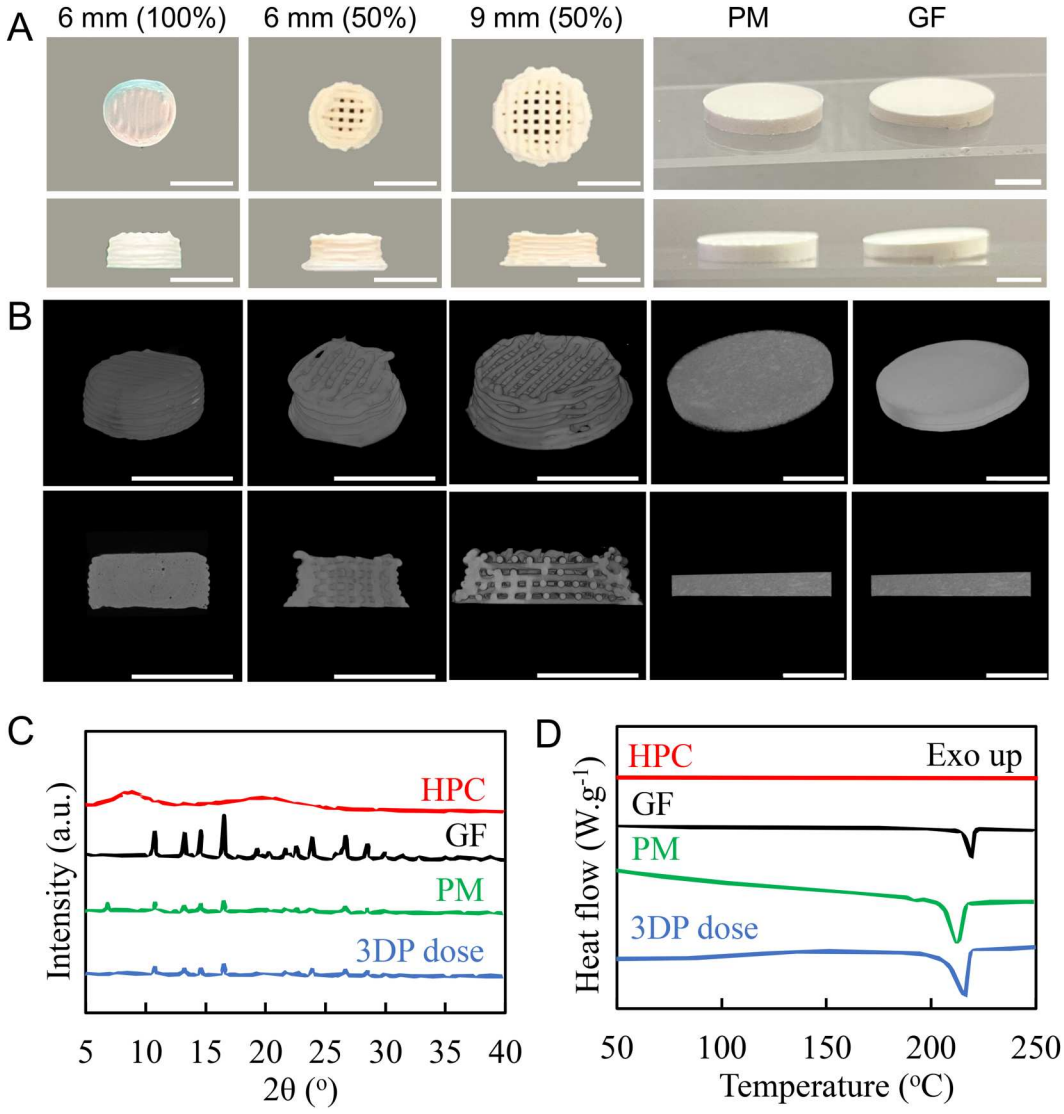

Figure 6. Imaging and characterisation of 3D-printed tablets alongside compressed powder mixture (PM) and griseofulvin (GF): (A) Pictures of 3D-printed tablets (using 40% ink) with 6 mm diameter (100% and 50% infill) and 9 mm diameter (50% infill), along with control PM and GF samples. (B) Corresponding µ-CT images (top: tilted view; bottom: cross-section). (C) XRD profiles of HPC, GF, PM, and 3D-printed dose. (D) DSC profiles for HPC, GF, PM, and 3D-printed dose. Scale bars = 5 mm.

conventional oral tablets, and for pediatric patients with a weak bite force, chewable gummies are approximately twice as soft as commonly used chewable tablets [36]. The stiffness (Young's Modulus, *E*) of the 3D-printed solid tablets was 74 ± 20 kPa, which is comparable to that of a commercial gummy (~10^2 kPa) (Figure 8A). When equilibrated in DI water (swollen) for 2 hours, the stiffness was further reduced to 23 ± 2 kPa. As expected, compressed tablets were significantly stiffer (~10^3 kPa) and not chewable. These findings indicate that the mechanical properties of our 3D-printed tablets align with commercial gummies for pediatric patients, supporting their classification as chewable, as suggested in the literature [30]. However, we acknowledge that this evaluation was limited to compressive modulus, and additional testing—including fracture force, toughness, and resilience—will be required to fully characterise chewable performance.

Tablet porosity is another key factor affecting both mechanical behaviour, stability (shelf-life), and drug release profile [93,94]. µCT imaging (Figure 6B) revealed that overall tablet porosity was low (1–1.5%) across all sample groups. However, 3D-printed solid tablets exhibited a slightly lower porosity (1.1%) compared to compressed tablets (1.5% for PM and 1.3% for GF).

Dissolution studies

In vitro drug dissolution tests were performed on 3D-printed tablets (40% water content) in a neutral/basic

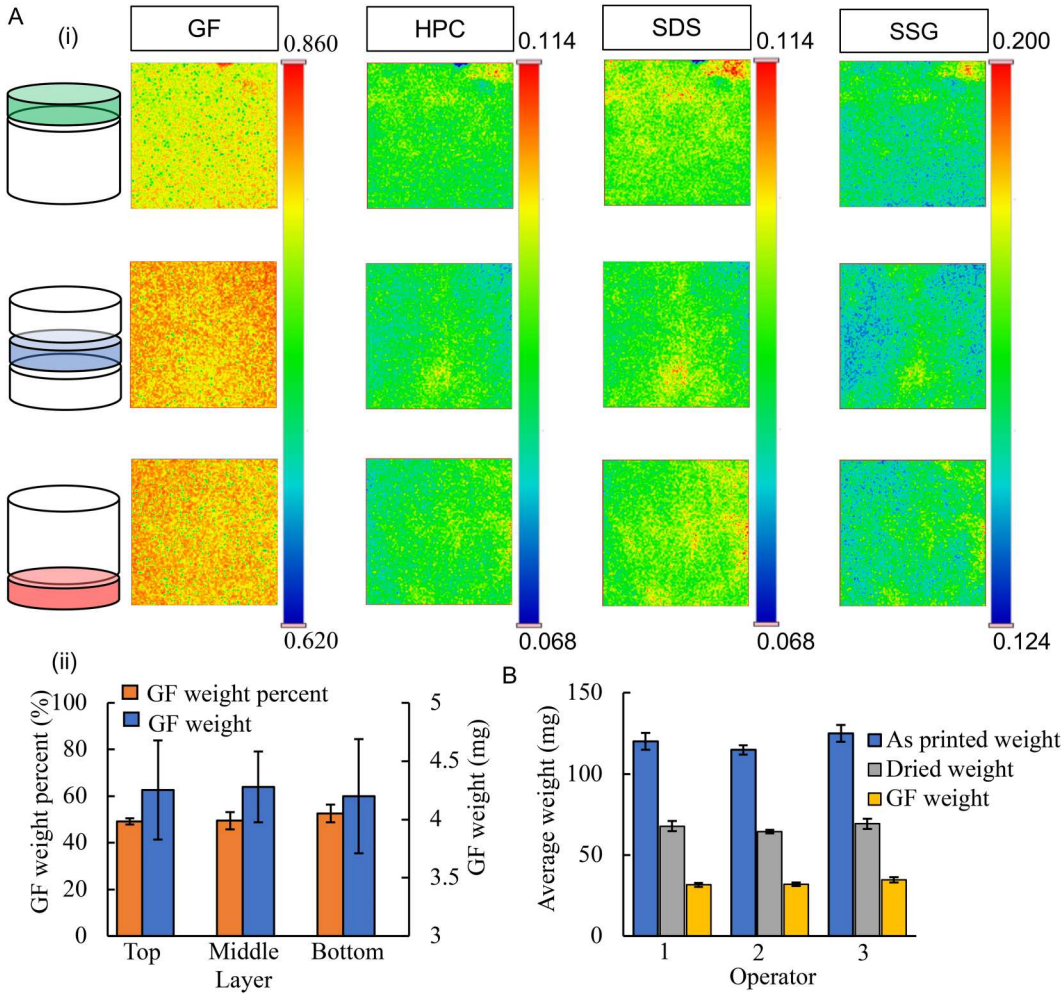

Figure 7. Intra-tablet homogeneity and operator effects on process quality: (A) Intra-tablet homogeneity assessed by (i) Raman mapping (50 × 50 µm) of 3D-printed tablets, showing the distributions of GF, HPC, SDS, and SSG across the bottom, centre, and top print layers, and (ii) UV-VIS analysis of GF content across the same layers. Data are presented as mean ± std (n = 3), with no statistical difference between layers. (B) As-printed weight, dried weight and GF weight (determined by UV-VIS) for 3D-printed tablets produced by operators with varying levels of expertise: Opr. 1 – An inexperienced undergraduate student, Opr. 2 – A graduate student with basic training, and Opr. 3 – A senior graduate student with extensive 3D printing expertise. Data are presented as mean ± std (n = 12), with no statistical difference between operators.

media, mimicking the intestinal fluid environment (Figure 8B). Due to the lipophilic nature of GF, it is expected to remain longer in the intestines than in the stomach [95]. This is important because dissolution media composition strongly influences API release profiles [96]. The 3D-printed tablets were designed as 3.2 mm tall cylinders with 6 mm diameter (100% and 50% infill) and 9 mm diameter (50% infill). The rationale for selecting 6 and 9 mm diameters with 50% infill is based on ensuring comparable drug content across different infill densities while maintaining a consistent total mass [25]. A 6 mm, 100% infill tablet has a similar mass to a 9 mm, 50% infill tablet, allowing for direct comparison of drug release profiles. Tablet weights were recorded to assess dosage uniformity, with average values of 0.12 ± 0.01 g (8.3% RSD) for 6 mm 100% infill tablets, 0.08 ± 0.007 g (8.8% RSD) for 6 mm 50% infill, and 0.14 ± 0.012 g (8.6% RSD) for 9 mm 50% infill, all falling within pharmacopeial limits for weight variation. The corresponding griseofulvin (GF) content was 32.5 ± 3.5 mg (10.8% RSD) for 6 mm, 100% infill tablets, 21.7 ± 2.3 mg (10.7% RSD) for 6 mm, 50% infill tablets, and 37.9 ± 4.1 mg (10.8% RSD) for 9 mm, 50% infill tablets, confirming consistent drug-to-weight ratios across all printed tablet designs. Additionally, these designs optimise the surface area-to-volume (SA/V) ratio, where 6 mm tablets represent a more compact geometry, while 9 mm tablets provide a larger surface area for dissolution, which may influence drug release kinetics.

To support this, SA/V ratios were calculated for each design: the 6 mm, 100% infill tablet had an SA/V of

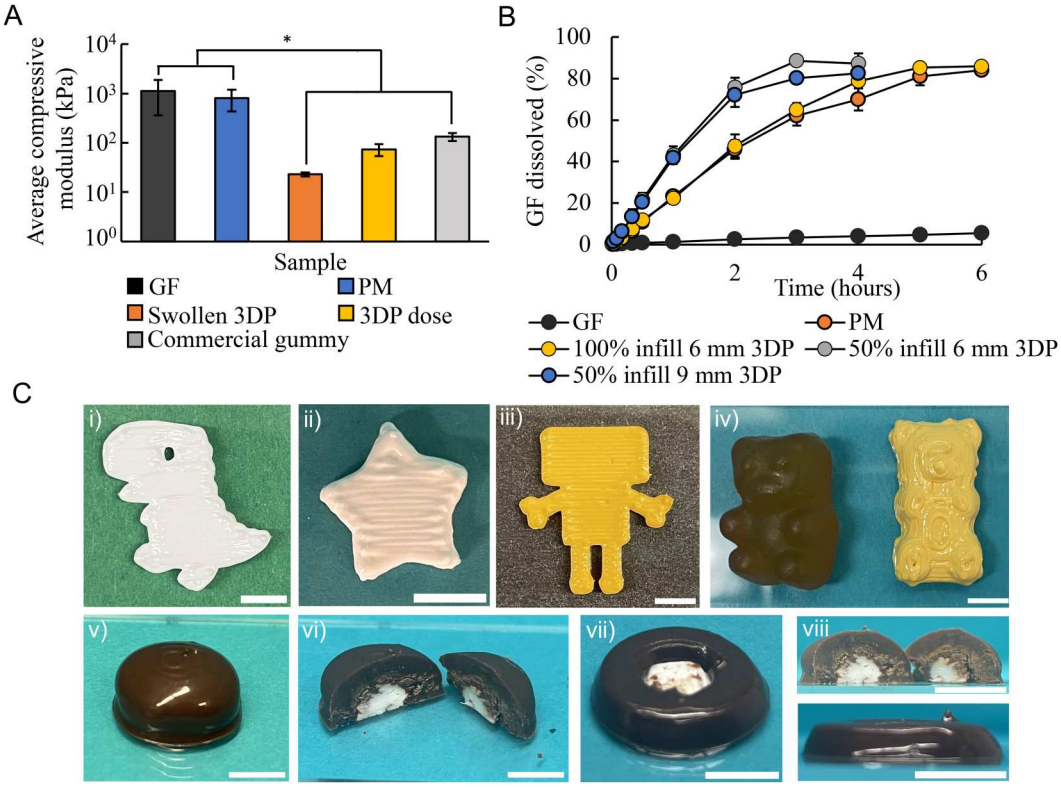

Figure 8. (A) Average compressive modulus of each sample, including 3D-printed (3DP) dose, swollen 3DP dose, compressed powder mixture (PM), GF powder, and a commercial gummy (Data are presented as mean ± std. for n = 3). (B) Percent dissolved GF over time for 3D-printed (3DP) tablets with 100% infill (6 mm diameter), 50% infill (6 and 9 mm diameter), as well as compressed PM and GF (Data are presented as mean ± std. for n = 3). (C) Images of 3D-printed tablets showcasing different shapes, colours, and compositions: (i–iv) single composition tablets without food colouring or with yellow food colouring, shaped as (i) dinosaur, (ii) star, (iii) robot, and (iv) bear (including a commercial gummy bear for comparison); (v–vii) single formulation tablets printed alongside chocolate, including (v) tablet, (vi) cross-section of a tablet showing chocolate-embedded drug, and (vii) donut with drug filling; (viii) cross-sectional views of tablets. Scale bars = 5 mm.

1.3 mm^{-1}, while the 6 and 9 mm, 50% infill tablets exhibited much higher SA/V values of 5.9 and 6.1 mm^{-1}, respectively. These findings reflect the increased surface exposure and porosity in 50% infill tablets and are consistent with the observed improvements in dissolution. The 50% infill tablets also introduce internal porosity, further affecting dissolution behaviour [26].

The dissolution profiles of 3D-printed and compressed tablets demonstrate the impact of tablet porosity, excipient composition, and structure on drug release. Compressed GF tablets exhibited minimal dissolution, confirming the poor solubility of GF in aqueous media. However, compressed PM tablets showed significantly enhanced dissolution, likely due to the presence of excipients facilitating drug dispersion. Among the 3D-printed formulations, reducing infill from 100% to 50% led to a faster release, indicating that porosity plays a crucial role in dissolution kinetics. The 80% release time for 100% infill 3D-printed tablets and compressed PM tablets was similar (~360 min), whereas 50% infill formulations reached 80% release in ~250 min. A similar trend was observed at the 40% release threshold, where 50% infill tablets released 40% of the drug in ~50 min, compared to ~120 min for 100% infill tablets. This suggests that introducing porosity significantly enhances dissolution by increasing fluid penetration and diffusion rates. This observation is consistent with prior studies demonstrating that increased porosity can accelerate drug release by facilitating media infiltration and expanding the effective surface area for dissolution. The positive correlation between porosity and dissolution rate has been reported across various drug classes with differing solubility profiles, further supporting the generalizability of this relationship [18,93,97,98]. While our study focused on a single model drug (griseofulvin), the observed trends align with the broader literature and reinforce the critical role of microstructural design in modulating drug release kinetics. The tablet size also played a role, as 9 mm, 50% infill tablets dissolved slightly slower than their 6 mm counterparts, suggesting that SA/V ratio contributes to dissolution kinetics. Despite these

variations in release rates, all formulations plateaued around 90–100% dissolution, indicating near-complete drug release. The dense structure of the 3D-printed tablets, potential hydrophobic interactions from GF, and dissolution medium limitations may contribute to the observed plateaus. Similar trends have been reported in previous studies on 3D-printed tablets, where incomplete dissolution was attributed to limited agitation and drug–polymer interactions [99–102]. Although media samples were collected at predefined intervals for UV-Vis analysis, real-time monitoring of pH or media composition was not performed. For BCS Class II drugs like GF, such in situ monitoring may provide additional insight into dissolution behaviour. Overall, these results highlight the importance of tuning porosity, excipient composition, and tablet size to optimise drug release without requiring changes to the ink formulation.

Custom tablet designs

One of the key advantages of 3D printing in pharmaceutical applications is its ability to create customised tablet designs with tailored shapes, porosity, and compositions. This capability enables precise control over drug release profiles, mechanical properties, and patient-specific formulations. Multi-material printing, which utilises multiple print heads dedicated to distinct ink formulations, allows for the fabrication of tablets with compositional hierarchy, enhancing both functionality and aesthetics.

In this study, we leveraged multi-material printing to fabricate tablets composed of multiple components, including a 40% griseofulvin nanosuspension ink, the same ink loaded with ~6% food dye for visual enhancement, and commercially available chocolate to improve palatability (Figure 8C). Two taste-masking tablet designs were explored: (1) a donut-shaped chocolate shell filled with the drug-loaded ink, and (2) a bulk chocolate tablet with the drug tablet embedded inside. In addition, a variety of child-friendly tablet shapes—including robots, dinosaurs, stars, and gummy bears—were printed using the dyed ink to assess colour uniformity and demonstrate the platform's potential for engaging pediatric patients. Although formal taste-masking and sensory evaluations were not conducted, these design strategies demonstrate the versatility of DIW-based 3D printing in producing customised, visually appealing, and palatable chewable dosage forms. While no drug–chocolate compatibility or stability studies were performed at this stage, these will be an important focus in future work to ensure chemical stability of the active pharmaceutical ingredient (API) during and after multi-material printing. Future studies will also evaluate the sensory performance and optimise these formulations to improve patient compliance, particularly in pediatric populations.

Beyond aesthetics and taste modifications, multi-material 3D printing holds significant potential for pharmaceutical applications, particularly in the development of complex dosage forms such as polypills [5]. By utilising separate syringes to print multiple formulations simultaneously, it is possible to create tablets with distinct compartments containing different active pharmaceutical ingredients (APIs), each with tailored release kinetics [23,24]. This approach enables personalised medicine by combining multiple therapies within a single oral dose, improving medication adherence and treatment efficacy.

Conclusions

In this study, we developed and characterised drug nanosuspension inks for the extrusion-based 3D printing of chewable oral tablets, establishing a scalable and adaptable platform that integrates formulation design, rheological profiling, printability evaluation, and predictive modelling. A key innovation of this work is the combination of rheology-informed formulation with machine learning (ML)-guided optimisation, enabling customisable tablet fabrication with high fidelity and reproducibility.

Ink formulation and rheology were systematically varied by adjusting water content (85–20 wt.%), revealing that shear-thinning behaviour, yield stress, and shear recovery are critical for maintaining extrudability and structural integrity. The 40% water formulation was identified as optimal, producing well-defined and self-supporting printed structures with minimal sagging or collapse. Quantitative and semi-quantitative printability assessments confirmed consistent performance under typical extrusion conditions.

Analytical modelling established predictive relationships between printed tablet diameter (D), rheological parameters, and printing inputs—specifically pressure (P), speed (v), and nozzle height (h). However, accuracy declined under extreme conditions (e.g. high P and low v), highlighting the need for a more robust predictive tool. To address this, we implemented machine learning models and found the Gradient Boosting Regressor (GBR) model yielded the highest accuracy ($R^2 = 0.94$, RMSE = 52 μm). The GBR model was further applied for inverse predictions to define optimal printing conditions based on target design outputs.

The printed tablets were thoroughly characterised. Micro-CT imaging demonstrated internal uniformity

and dimensional precision. XRD and DSC confirmed that GF remained in its crystalline form post-processing, preserving its physical stability. Raman mapping and UV-VIS spectroscopy demonstrated high drug content uniformity, with negligible operator-dependent variability, reinforcing the reproducibility and robustness of the manufacturing process.

Mechanical testing showed that the 3D-printed tablets had a Young's modulus (~74 kPa) comparable to commercial gummies, and further softened in aqueous conditions, meeting mechanical criteria for pediatric chewable formulations. Dissolution testing in simulated intestinal fluid confirmed that tablet porosity and infill density modulated drug release kinetics, enabling customisation of release profiles without changing the ink formulation.

Finally, the versatility of the platform was demonstrated through multi-material printing, incorporating excipients like food dyes and chocolate for enhanced palatability. The potential for spatially controlled multi-compartment polypill fabrication was also discussed, positioning this approach as a promising tool for personalised oral drug delivery. In summary, this study establishes a comprehensive workflow for 3D printing chewable tablets, from ink formulation and rheological characterisation to printability optimisation, ML-enhanced predictability, and tablet characterisation. By integrating rheology-based analytical modelling with machine learning, we present a scalable and adaptable platform for the fabrication of customisable, patient-specific dosage forms.

While the current model was developed using a single ink formulation, it serves as a foundational step toward predictive printing. We acknowledge that rheological and compositional variables were not included in the current ML inputs, which limits broader generalizability. Ongoing work is focused on expanding the model to incorporate formulation-dependent parameters and validating its predictive performance across multiple ink batches. Future efforts will also explore multi-drug polypill designs and further refine the framework for personalised treatment strategies.

It is important to note that this study primarily focused on printability and formulation-process relationships, and does not comprehensively address all critical quality attributes required for chewable oral tablets intended for human use, as outlined in the FDA's *Quality Attribute Considerations for Chewable Tablets* guidance (CDER, August 2018). Key parameters such as tablet hardness and friability, disintegration time, and performance in simulated saliva were not evaluated. These represent important limitations of the current work and will be addressed in future studies to align the platform with regulatory standards and advance its translational potential.

Experimental methods

Preparation of drug nanosuspension ink formulations: The drug suspension composed of micronized griseofulvin (GF, BP/EP grade, Letco Medical, Decatur, AL, USA), sodium dodecyl sulfate (SDS, ACS grade, GFS chemicals, Columbus, OH, USA), hydroxypropyl cellulose (HPC, L grade, Nisso America, New York, NY, USA), sodium starch glycolate (SSG, Primojel PhC, DFE Pharma, Foxhol, The Netherlands), and deionized (DI) water prepared by wet stirred media milling (WSMM, MicroCer, NETZSCH Fine Particle Technology LLC, Exton, PA, USA). All ingredients were used as received except for SSG, which was sieved (US Standard Testing Sieves, ASTM E-11) using a sieve shaker (Octagon 2000, Endecotts Ltd.) for 1 h to collect particles < 38 µm. A pre-suspension containing 20 g GF, 10 g HPC, and 1 g SDS in 200 g DI water was prepared under constant shear mixing (Cat#. 14-503, Fisher Scientific, Pittsburgh, PA, USA) for 2 h at 300 rpm. The pre-suspension was transferred into the mill along with 50 mL bulk volume of Yttria-stabilized zirconia beads (400 mm in diameter) as milling media. 3 g SSG was added gradually within the first 12 min of milling to prevent clogging with swollen SSG particles as per previous protocol [47]. The total milling time was 90 min at 3200 rpm at a controlled temperature below 34°C. The particle size distribution (PSD) of the drug nanosuspension was characterised by laser diffraction (Coulter LS 13 320, Beckman Coulter, Miami, FL, USA). The diluted samples were added dropwise to de-ionized water until the polarisation intensity differential scattering (PIDS) reached above 40%. The PSD was computed using Mie theory and in-built software. SEM imaging of the samples was carried out with the JSM7900F Schottky Field Emission Scanning Electron Microscope (JEOL) equipped with both secondary and backscattered electron detectors at 10 kV operating voltage. Transission electron microscopy (TEM) was performed on a JEM-F200 at 200.00 kV operating voltage. Samples were fixed on a 400 mesh copper support film. Dilute sample was transferred onto carbon tape dropwise then sputter coated (Q150 T, Quorum Technologies Ltd, East Sussex, England) with 5.0–15.9 nm 0.80Au/0.20Pd to prevent charging during imaging.

In addition to as prepared ink formulation with 85% water, formulations with 70, 60, 40, and 20% water was prepared by evaporating the DI water at 35°C. For this purpose, nanosuspensions (15 g batches) were transferred into polystyrene petri dishes (100 × 15 mm)

and kept in a convection oven for up to 700 min. The decrease in the weight of the dishes was recorded with time with respect to initial weight to calculate the water evaporation (mass reduction) rate. Samples (3 samples per dish) were collected during evaporation to confirm the composition of the inks using TGA (TGA/DSC1 STAR e System, Mettler Toledo Inc., Columbus, OH). Each sample was placed in a ceramic crucible and heated at a rate of 10°C·min^{-1} under 60 mL·min^{-1} nitrogen flow from 25–150°C.

Rheology of the ink formulations: Rheology of the ink formulations were characterised using a Kinexus Ultra + Rheometer (Netzsch) with parallel plate geometry. All measurements were performed using a Kinexus Peltier disposable bottom plate and Malvern reusable 20 mm upper plate (Malvern) using approximately 0.24 mL of ink at 25°C at a fixed (750 µm) gap. Painter's tape or sandpaper was used to prevent slipping. Ink viscosity was measured using a shear ramp from 0.01–1,000 s^{-1}. Elastic modulus (G') and viscous modulus (G'') were measured using a strain sweep from 0.1–300% at 1 Hz to determine the yield stress ($G' = G''$) of each ink formulation. Frequency sweep tests (0.1–100 Hz) were performed at 0.05% strain to monitor the change in G' and G''. The shear recovery behaviour of the inks was evaluated by monitoring the change in shear moduli over time while alternating between low (0.05%) and high (300%) strain. The high shear rate at the nozzle wall during printing was calculated using the Rabinowitsch-Mooney equation:

$$\dot{\gamma}_{avg} = \left(\frac{3 + \frac{1}{n}}{4}\right)\left(\frac{4Q}{\pi R^3}\right) \tag{5}$$

where n is a material-specific parameter, R is the outlet radius of the tapered nozzle, and Q is the volumetric flow rate [103]. Q was obtained using a previously developed protocol where density is used to convert a measured gravimetric flow rate to volumetric flow rate [70]. Briefly, gravimetric flow rate was measured by extruding ink onto a plate for a known amount of time using 'purge' setting while density is measured by measuring the weight of a printed cylinder with known dimensions [70].

Machine Learning: The dataset used for predicting line width based on nozzle offset, pressure, and printing speed was split into training (75%) and testing (25%) sets using the Train Test Split function from the Model Selection module of the Sklearn Python library. The Standard Scaler object from Sklearn's preprocessing module was applied to standardise both the training and testing variables for input features and target values. In this study, Principal Component Analysis (PCA) was applied to reduce the dimensionality of the input features, transforming the scaled training and testing data into three principal components, which captured the most significant variance in the dataset for model training and evaluation. To ensure model robustness, a 10-fold cross-validation was performed using KFold (n splits = 10, shuffle = True, random state = 42) from Model Selection. For curve fitting and prediction, four regression models were used: Linear Regressor, Random Forest Regressor, Gradient Boosting Regressor, and KNeighbors Regressor (Sklearn). The Linear Regression model was used with default settings, and predictions for both training and test data were plotted to visually assess performance. For the Random Forest Regressor, the model was initially used with the default setting of n_estimators = 100. Hyperparameter tuning was then performed using GridSearchCV over a range of max_depth values, with the optimal configuration of n_estimators = 150 and max_depth = 3 selected for further evaluation. The Gradient Boosting Regressor was tuned using RandomizedSearchCV to optimise hyperparameters such as n_estimators, learning_rate, max_depth, and others, with the best configuration identified as n_estimators = 1350, criterion = 'friedman_mse', max_leaf_nodes = 2, subsample = 0.8, learning_rate = 0.2, and max_depth = 2. Finally, the KNeighbors Regressor was used with the default setting of n_neighbors = 5. Detailed hyperparameter tuning processes, including the grids used for GridSearchCV and RandomizedSearchCV, are provided for reproducibility.

The predictions of the test data were visualised using scatter plots generated by the Matplotlib library. The performance of each model was assessed using root mean square error (RMSE), and R^2 score to evaluate prediction reliability. After selecting the best-performing model, it was used for inverse prediction. The process utilised the minimise function from the SciPy library, employing the Nelder-Mead method to iteratively adjust the input parameters. The objective function was defined to minimise the absolute difference between the predicted and target line width. To ensure consistency with the trained model, the input values were first standardised using the previously fitted scaler and then transformed using the trained PCA model before being passed to the Gradient Boosting Regressor for prediction. By optimising the objective function, the model identified the optimal nozzle offset, pressure, and printing speed required to achieve the desired line width. After obtaining the parameters for the desired line width, experiments were conducted using these parameters, and the actual line width was measured. This measured line width was then compared

to the desired value to verify the model's effectiveness for practical application of printing conditions.

3D Printing of tablets: 3D Bioplotter (Starter Series, EnvisionTEC) was used for 3D printing. 3 mL syringes (Nordson EFD Inc.) with a 22-gauge (410 µm in diameter) tapered dispense tip (Fisnar) were used. All samples were printed at room temperature onto glass slides. All digital designs were created by using Fusion 360 (Autodesk) and sliced by Rapid Prototyping (RP) software with a slicing layer thickness of 400 µm. The optimised printing parameters for each ink formulation with 400 µm offset were determined by printing with the grid pattern, with nozzle speed and applied pressure varying based on ink formulation. For 40%, the speed is 5 mm/s at 600 kPa; for 60%, it's 4 mm/s at 100 kPa; for 70%, it's 10 mm/s at 20 kPa; and for 85%, the original formulation, the speed is 35 mm/s at 10 kPa. Tablets were designed as 3.2 mm tall cylinders with 6 mm (100 or 50% infill) or 9 mm in diameter (50% infill). Grid patterns composed of 550 µm struts (in diameter) with a 1 mm strut-to-strut spacing were printed to evaluate the printability (print fidelity). Nozzle speed and pressure were adjusted for each ink formulation to achieve 550 µm struts. Continuous struts (lines) were printed using 400 µm offset (needle to print surface distance determined by the needle size) to determine the effect of print pressure (4–5 bar) and speed (3–15 $mm \cdot s^{-1}$) on strut diameter (or width). This study was repeated multiple times with different batches of ink to test the reproducibility of the data. At least 15 images were collected from different regions of each strut for analysis. Images were taken using a Nikon Eclipse Ti-S optical microscope fitted with a Nikon Digital Sight DS-Ri1 camera. ImageJ (NIH) was used for image analysis.

Characterization of tablets: 3D-printed tablets were characterised using XRD (PANanalytical) and DSC (Mettler–Toledo DSC 3) to investigate the drug crystallinity. As received GF, HPC, and a physical mixture (PM) of as-received powder excipients in the same ratio as the suspension (PM, 59 wt% GF, 29 wt% HPC, 3 wt% SDS, 9 wt% SSG) were used as controls. For XRD, dried tablets were grinded with mortar and pestle for 30 min to obtain a powder then scanned from 5–40° at a rate of 0.165 s^{-1} for the 2θ scanning mode equipped with Cu Kα radiation (λ = 1.5406 Å). For DSC, 6–7 mg of dry sample was placed in a 40 µL sealed perforated aluminum pan and heated from 25–250°C at a rate of 10°C·min^{-1} with nitrogen flow of 60 mL·min^{-1}. Data analysis was performed using STARe software (V16.20, Mettler–Toledo).

A Kinexus Ultra + Rheometer (Netzsch) was used to perform uniaxial compression tests (sample size ≥ 3) to determine the Young's modulus (*E*) of the printed tablets. For this purpose, tablets were printed using the 40% formulation and tests were performed before and after swelling in DI water for 2 h. Compressed tablets and commercially available pediatric gummies (Equate Children's Multivitamin Gummies, Walmart) were used as controls. The compressed tablets contained either as-recieved GF only or the aforementioned PM of powder excipients. GF and PM tablets were compacted (Gamlen D1000, Gamlen Tableting Limited, Beckenham, UK) in 13 mm x 3 mm dies and weighed 300 mg each. Swelling ratio of the 3DP tablets was calculated at 0.5, 1, and 2 h incubation in DI water using the following equation:

$$S = \frac{w_d - w_s}{w_d} \cdot 100 \tag{6}$$

where S is the swelling ratio, w_d is the dry weight, and w_s is the swollen weight.

A micro-computed tomography scanner (µ-CT, SkyScan 1275, Bruker) was used to investigate the architectural uniformity and porosity of the tablets. Compressed PM and GF only tablets were used as controls. Compositional uniformity was characterised by Raman imaging microscope (DXR2xi, Thermo Fisher Scientific Inc.) using a DXR 532 nm filter and 50X objective lens with 10 scans at 0.05 s·$scan^{-1}$ at 0.5 µm per pixel resolution using 10 mW laser power. Scanning region was 50 × 50 µm square for each sample. To isolate individual layers for analysis, a single batch of ink was used, and each print layer (bottom, centre, and top) was printed separately on a flat substrate. Specifically, the standard tablet design was used, but the printing process was paused after each layer: after printing the bottom layer, the print job was stopped and the layer was isolated; printing then resumed on a new surface to generate the centre layer, and finally the top layer was printed and isolated in the same manner. This approach allowed for layer-specific compositional analysis while maintaining consistent formulation and printing conditions across all layers. Data acquisition, image processing, and correlation index facilitated by OMNICxi Raman Imaging Software (Thermo Fisher Scientific Inc.). All correlation indices were generated using reference spectra of pure materials at identical imaging conditions.

3DP tablets for characterisation were always printed using 40% ink. Operators would adjust the parameters to get the best quality print possible based on that ink's properties, changing print parameters (*e.g.* 5–6 bar, 3–20 mms^{-1}) based on inter-batch inhomogeneities. After printing, tablets were dried in the oven at 35°C for about 2 h. Tablets used for characterisation were printed with the following cylindrical dimensions

(diameter x height) and infills: 6 × 3.2 mm, 100% infill; 6 × 3.2 mm, 50% infill; or 9 × 3.2 mm, 50% infill.

Tablet Dissolution and Drug Assay: Tablet dissolution was investigated using a Distek 2100C dissolution tester following the USP II paddle method using 1000 mL of 3 gL^{-1} SDS-deionised water solution at 37° C at 50 rpm paddle speed. 3 gL^{-1} SDS solution was selected as the dissolution medium as it provided good discrimination of the samples. The drug content in the tablets was found after determining the moisture content via TGA. For each tablet, 4 mL of sample was collected at 1, 2, 5, 10, 20, 30, 60, 120, 180, 240, 300 and 360 min. Porous tablets were measured until 240 min. as they reached equilibrium. Collected samples were filtered through 0.1 µm polyvinylidene fluoride (PVDF) membrane-type syringe filter before ultraviolet (UV) spectroscopy measurements. 0.2 mL of the filtered samples were diluted with 2.5 mL of 37°C dissolution medium before UV measurement. Dissolved GF amount was measured by UV spectroscopy at 292 nm wavelength and calculated using a pre-established calibration curve. 3 gL^{-1} SDS solution was used as the blank before UV measurement and 3 replicates of each sample were performed.

Statistical analysis, curve fitting: All statistical analyses and regressions were performed using Minitab® software (Version 21.2). All data is presented as mean ± standard deviation unless otherwise specified, and statistical comparisons performed using two-way ANOVA with post-hoc analysis. All percentages are weight per total weight unless otherwise specified.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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Data availability statement

The data that support the findings of this study are available online (https://doi.org/10.6084/m9.figshare.28716182.v1).

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